

AP Project

Sample data:

//admin

Username- admin1

Password - \$2a\$10\$N9qo8uLOickgx2ZMRZoMye.KbJ1Q.2V.8.5.5J5J5J5J5J5J5J

// instructor

Username - inst1

Password - \$2a\$10\$N9qo8uLOickgx2ZMRZoMye.KbJ1Q.2V.8.5.5J5J5J5J5J5J5J

//student

Username - student1

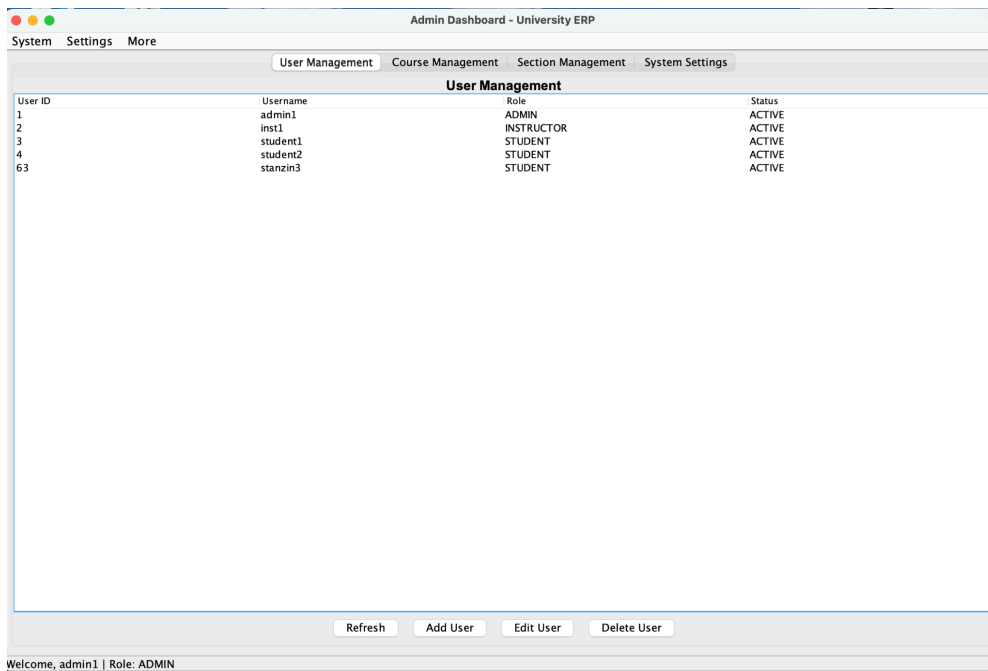
Password - \$2a\$10\$N9qo8uLOickgx2ZMRZoMye.KbJ1Q.2V.8.5.5J5J5J5J5J5J5J

Username- student2

Password - \$2a\$10\$N9qo8uLOickgx2ZMRZoMye.KbJ1Q.2V.8.5.5J5J5J5J5J5J5J

Screenshot and reports:

1) Admin Dashboard

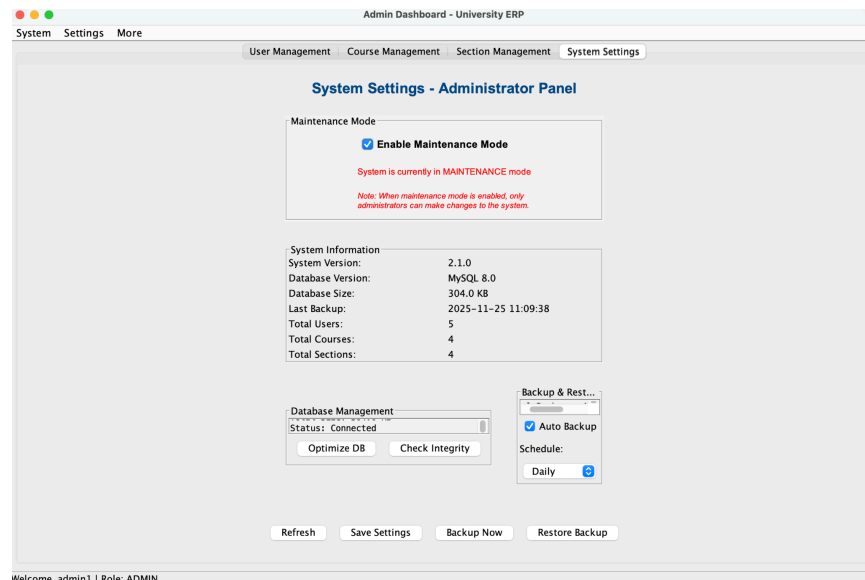


The screenshot displays the 'Admin Dashboard - University ERP' interface. At the top, there are navigation tabs: 'System', 'Settings', and 'More'. Below these, a sub-navigation bar includes 'User Management', 'Course Management', 'Section Management', and 'System Settings'. The 'User Management' section is active, showing a table with the following data:

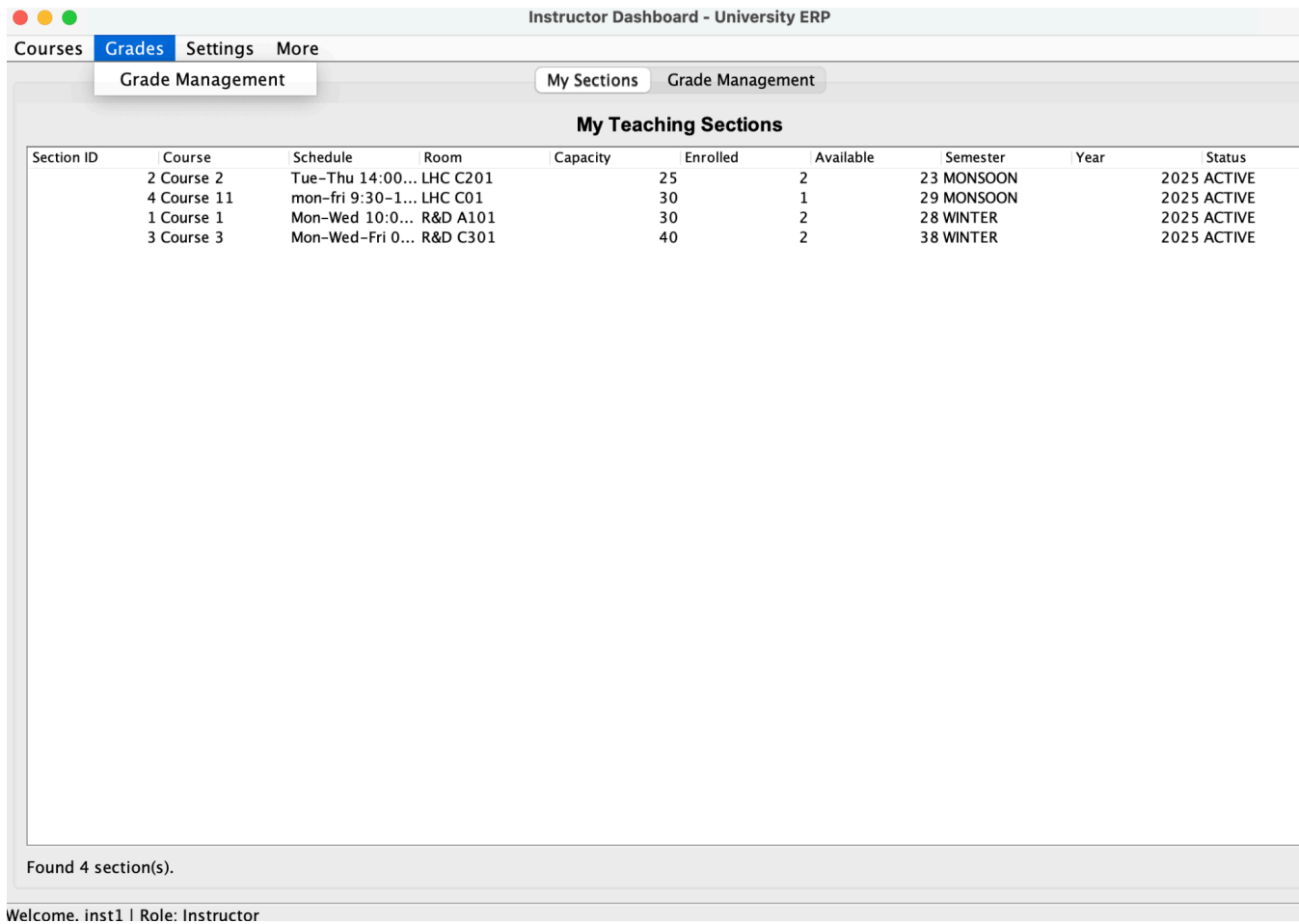
User ID	Username	Role	Status
1	admin1	ADMIN	ACTIVE
2	inst1	INSTRUCTOR	ACTIVE
3	student1	STUDENT	ACTIVE
4	student2	STUDENT	ACTIVE
63	stanzin3	STUDENT	ACTIVE

At the bottom of the table, there are four buttons: 'Refresh', 'Add User', 'Edit User', and 'Delete User'. The footer of the dashboard reads 'Welcome, admin1 | Role: ADMIN'.

- With maintenance mode



2) Instructor Dashboard



- With maintenance mode

The screenshot shows the 'Instructor Dashboard - University ERP' interface. At the top, there are navigation links: 'Courses', 'Grades', 'Settings', and 'More'. A prominent red banner across the top of the main content area reads 'MAINTENANCE MODE - Read Only'. Below this banner, there are two tabs: 'My Sections' (which is selected) and 'Grade Management'. Under the 'My Sections' tab, the heading 'My Teaching Sections' is displayed above a table. The table lists four sections with columns for Section ID, Course, Schedule, Room, Capacity, Enrolled, Available, Semester, Year, and Status. At the bottom of the interface, a status message says 'Found 4 section(s)'.

Section ID	Course	Schedule	Room	Capacity	Enrolled	Available	Semester	Year	Status
2	Course 2	Tue-Thu 14:00-1...	LHC C201	25	2	2	23 MONSOON		2025 ACTIVE
4	Course 11	mon-fri 9:30-1...	LHC C01	30	1	1	29 MONSOON		2025 ACTIVE
1	Course 1	Mon-Wed 10:00-...	R&D A101	30	2	2	28 WINTER		2025 ACTIVE
3	Course 3	Mon-Wed-Fri 0...	R&D C301	40	2	2	38 WINTER		2025 ACTIVE

3) Student Dashboard

The screenshot shows the 'Student Dashboard - University ERP' interface. At the top, there are navigation links: 'Academic', 'Settings', and 'More'. Below these, there are three tabs: 'My Registrations' (selected), 'Course Catalog', and 'Grades'. Under the 'My Registrations' tab, the heading 'My Course Registrations' is displayed above a table. The table lists four registrations with columns for Enrollment ID, Course Code, Course Title, Section, Status, and Enrolled Date. A modal window titled 'Course Catalog - Register New Course' is open in the foreground, showing a list of courses with columns for Course Code, Title, and Credits. At the bottom of the modal, there are two buttons: 'View Sections' and 'Close'. At the bottom of the main dashboard, there are three buttons: 'Refresh', 'Drop Course', and 'Register New Course'.

Enrollment ID	Course Code	Course Title	Section	Status	Enrolled Date
20	CSE205	Machine Learning	4	REGISTERED	2025-11-26 02:00:38.0
19	CSE201	Data Structures	2	REGISTERED	2025-11-26 01:56:30.0
12	CSE202	Advanced Programming	3	REGISTERED	2025-11-25 20:13:18.0
1	CSE101	Intro to Programming	1	REGISTERED	2025-11-25 09:18:44.0

Course Code	Title	Credits
CSE101	Intro to Programming	4
CSE201	Data Structures	4
CSE202	Advanced Programming	4
CSE205	Machine Learning	4

- With maintenance mode

Student Dashboard - University ERP

Academic Settings More

MAINTENANCE MODE - Read Only

My Registrations Course Catalog Grades

MAINTENANCE MODE: Course registration/drop disabled

Enrollment ID	Course Code	Course Title	Section	Status	Enrolled Date
20	CSE205	Machine Learning	4	REGISTERED	2025-11-26 02:00:38.0
19	CSE201	Data Structures	2	REGISTERED	2025-11-26 01:56:30.0
12	CSE202	Advanced Programming	3	REGISTERED	2025-11-25 20:13:18.0
1	CSE101	Intro to Programming	1	REGISTERED	2025-11-25 09:18:44.0

Refresh Drop Course (Disabled - Maintenance) Register (Disabled - Maintenance)

Welcome, student1 | Role: Student | **MAINTENANCE MODE - Read Only**

Test Cases & Results:

- Login ;Incorrect password message shown:

Login – Valid Credentials :

ERP System - Login

ERP

Username:

Password:

LOGIN

admin1 | inst1 | student1 | student2

Login – invalid Credentials

ERP System - Login

ERP

Username:

Password:

LOGIN

Incorrect username or password.

admin1 | inst1 | student1 | student2

- Password Management;

Change Password – Strength Validation:

All validations triggered correctly, and the correct error message was shown for short/weak/good passwords. Password update succeeded only with valid inputs.

```
se 11    mon-fri 9:30-1... LHC C01      30      1      29 MONSOON
se 1     Mon-Wed 10:0... R&D A101     30      2      28 WINTER
se 3     Mon-Wed-Fri 0... R&D C301    40      2      38 WINTER
```

Change Password

Change Your Password

Password Change

Current Password:

New Password:

Strength: Weak

Confirm Password:

Requirements: At least 8 characters, 1 uppercase, 1 digit

Change Password Cancel

Change Password

Change Your Password

Password Change

Current Password:

New Password:

Strength: Fair

Confirm Password:

Requirements: At least 8 characters, 1 uppercase, 1 digit

Change Password Cancel

Change Password

Change Your Password

Password Change

Current Password:

New Password:

Strength: Good

Confirm Password:

Requirements: At least 8 characters, 1 uppercase, 1 digit

Change Password Cancel

- Registration Failure Handling;

Trying to register for a course the student is already enrolled in / section full, proper message displayed.

Student Dashboard - University ERP

Academic Settings More

My Registrations Course Catalog Grades

Course Catalog

Course Code	Title	Credits	Description	Status
CSE101	Intro to Programming	4		ACTIVE
CSE201	Data Structures	4		ACTIVE
CSE202	Advanced Programming	4		ACTIVE
CSE205				

Sections for: Data Structures

Section ID	Instructor	Day/Time	Room	Capacity	Available Seats	Semester
2	Instructor	Tue-Thu 14:0...	LHC C201	25	23	MONSOON 20...

Registration Failed

Failed to register for section. It may be full or you may already be registered.

OK

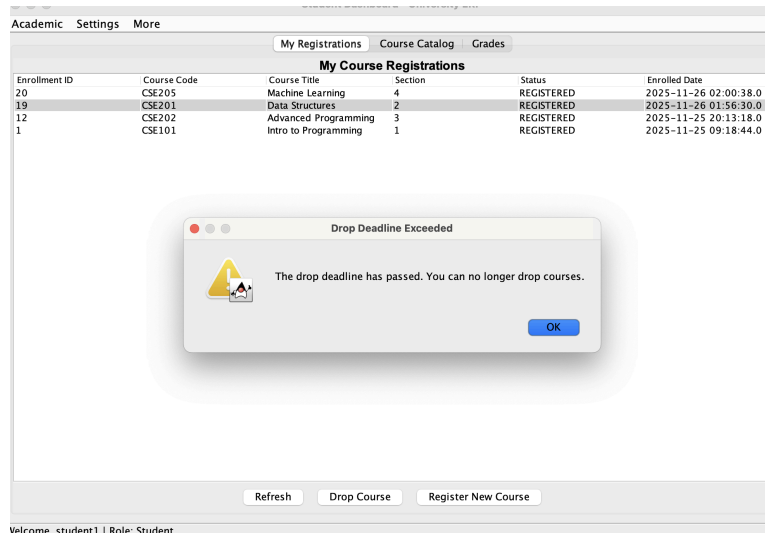
Register for Selected Section Close

Refresh View Sections Register for Course

Welcome: student1 | Data Structures

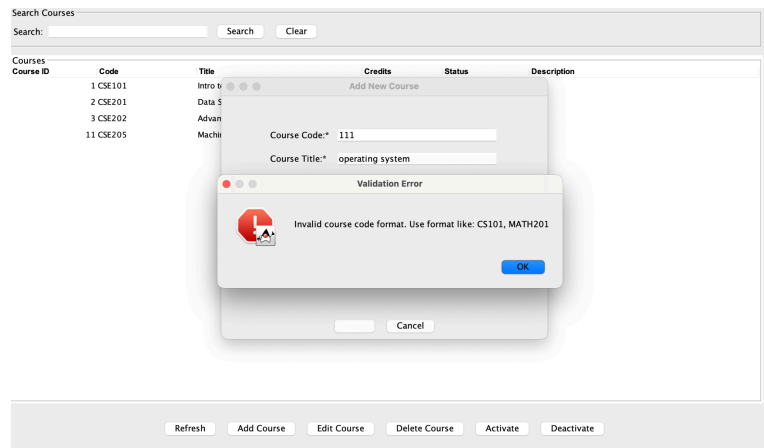
- Drop Deadline Enforcement:

Student attempts to drop a course after deadline ,drop option is blocked and show clear message for the same .



- Invalid Course Code Format;

Admin tries to add a course using an incorrect format .



Other properties:

This University ERP system was built and tested using a standard **Java + MySQL + Maven** setup. The backend follows a modular architecture with DAO, service, and UI layers, while MySQL stores all authentication and ERP data.

- Databases & Tables:

```
## Database 1: university_auth_db
```

user_id	username	role	password_hash	status	failed_attempts	locked_until	created_at
1	admin1	ADMIN	[hash]	ACTIVE	0	NULL	2025-11-27 10:00
2	inst1	INSTRUCTOR	[hash]	ACTIVE	0	NULL	2025-11-27 10:00
3	stu1	STUDENT	[hash]	ACTIVE	0	NULL	2025-11-27 10:00
4	stu2	STUDENT	[hash]	ACTIVE	0	NULL	2025-11-27 10:00

Tables_in_university_erp_...	
courses	
enrollments	
grades	
instructors	
sections	
settings	
students	
users	

```
## Database 2: university_erp_db
```

```
### Table: courses
```

course_id	code	title	credits
1	CSE101	Intro to Programming	3
2	CSE102	Data Structures	4
3	MTH201	Calculus I	4

```
### Table: instructors
```

user_id	department	office
2	Computer Science	CS-201
5	Mathematics	M-105
6	Computer Science	CS-305

```
## Table: settings
```

id	key_name	value	description
1	maintenance_mode	false	System maintenance flag
2	drop_deadline	2025-12-31	Last date to drop courses
3	semester	Fall 2025	Current semester
4	academic_year	2025	Current academic year
5	max_courses	5	Max courses per student per semester

```
## Table: grades
```

grade_id	enrollment_id	assign_w	midterm_w	final_w	assign_s	midterm_s	final_s	weighted_score	letter
1	1	30	30	40	85	78	82	81.6	B
2	2	25	35	40	88	92	98	98.2	A
3	3	30	30	40	75	70	68	78.8	C
4	5	30	30	40	92	95	98	95.2	A

```
### Table: enrollments
```

enrollment_id	student_id	section_id	status	enrolled_at
1	3	1	REGISTERED	2025-11-01 09:30
2	3	3	REGISTERED	2025-11-01 09:35
3	4	1	REGISTERED	2025-11-01 09:40
4	4	2	DROPPED	2025-11-25 14:15
5	7	1	REGISTERED	2025-11-05 10:00

```
## Table: sections
```

section_id	course_id	instructor_id	day_time	room	capacity	semester	year
1	1	2	Mon-Wed 10:00-11:30	A-101	30	Fall 2025	2025
2	1	2	Tue-Thu 14:00-15:30	A-102	25	Fall 2025	2025
3	2	5	Mon-Wed 13:00-14:30	M-201	35	Fall 2025	2025

- Maven Project Configuration:

The project uses **Maven** to manage building and packaging the whole application.

Maven Compiler Plugin :This makes sure the project is compiled using **Java 21**, so everything is built with the correct language version.The shade plugin ensures the final JAR contains all required classes, simplifying execution.

- Project Dependencies:

The application relies on the following libraries:

Security & JWT- Used for secure token generation and request validation.

Database - Provides the JDBC driver for MySQL database access.

Servlet API - Used for UI dialogs and handling event logic.

Conclusion :

Overall, the University ERP system functioned well for all the main features we tested. The database tables, Maven setup, dependencies, and build process all supported the application without any problems. Key actions such as login, registration, password changes, GPA checks, and admin validations performed as expected. The system managed errors and constraints, like deadlines and invalid inputs, effectively.